



AI reinvention made real

Accenture Innovation Challenge 2026



SAMOSA DRIVEN DEVELOPMENT

Problem Statement PatientTriage.ai

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THE CRISIS IN INDIAN MEDICAL EMERGENCY DEPARTMENTS

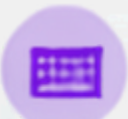


High Stakes Under Extreme Pressure

Emergency Departments(EDs) face unprecedented surges, leading to catastrophic delays. The critical decision of patient sequencing falls on triage nurses working under **immense cognitive load, extreme pressure, and severe time constraints**. Mis-priorization costs lives. Our Goal is to optimize routing without replacing legally required human clinical judgment.

Core System Constraints

- ⚠ The Boundary of AI(Decision vs Recommendation): Must strictly separate autonomous decisions from recommendations requiring human approval.
- ⚠ The 90-Second Window: System must rely only on data seamlessly gathered within the first 90 seconds of arrival.
- ⚠ Designing for Extremes: Must fail safely under worst-case conditions.



Data & Intake

- **Data Entry Bottleneck**
No time for typing during active bleeding or cardiac events.
- **Unidentified Arrival**
Unconscious patients with no ID or family present on scene.
- **Language Barriers**
Regional dialects in metro and tourist cities create noisy data.
- **Hidden Symptoms**
Patients may not disclose critical details unless specifically prompted.
- **Incomplete Histories**
The system must still work with partial intake data.



Clinical & Demo. Bias

- **Waiting Room Decay**
Patients marked stable silently deteriorate while waiting, with no automatic refrage trigger.
- **Age-Based Blindspots**
AI trained on adult data misclassifies children and elderly whose normal vitals differ drastically.
- **Cultural Bias**
Pain levels underestimated due to demographic stoicism, skewing urgency scores donnward.



Flow & Crowding

- **“Frequent Flyer” Overcrowding**
Minor ailments(like mild colds or simple rashes) flood ERs, delaying actual emergencies.
- **Resource Blindspots**
Lack of real-time mapping for available doctors,beds or nearby alternative clinics at 150% capacity.
- **Mass Casualty Surges**
Unpredictable quantity of patients during events like a major bus accident overwhelms standard queues.



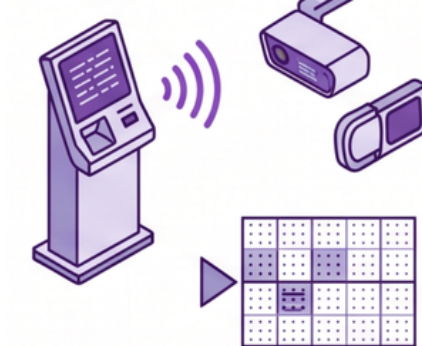
Environmental Chaos

- **Infrastructure Unreliability**
Wi-Fi drops, server lags, and power outages in hospitals is an unsaid Reality.
- **Nurse Cognitive Overload**
Attentional tunneling from noise, clutter in ER, and constant alarms.
- **Alert Fatigue**
Overwhelmed staff may miss data or ignore AI entirely and get biased over decisions.



NEXT-GEN AI TRIAGE SUPPORT SYSTEM

Transforming emergency triage into a dynamic, continuous workflow via a strict "Human-in-the-Loop" architecture.



Phase 1: Rapid Intake (< 90s)

- **Multilingual NLP Kiosk:** Uses acoustic stress detection to bypass language barriers and identify extreme distress.
- **Vision-First IoT:** Overhead cameras and instant-clip SpO2 sensors instantly capture trauma and vitals silently.
- **Smart Baselines:** AI asks targeted follow-ups and applies age-calibrated thresholds to prevent algorithmic blindspots.



Phase 2: Core Processing Engine

- **Autonomous Routing:** Independently identifies low-risk cases, routing them to fast-track units to decongest the ER.
- **Advisory Validation:** High-acuity recommendations require mandatory, manual nurse sign-off ("Accept/Reject").
- **Fail-Safe Architecture:** AI is structurally locked out from autonomously lowering priority levels without human authentication.



Phase 3: Continuous Monitoring

- **Supply Chain Routing:** The queue is simulated and re-sorted every 60s via a "clinical decay curve" to prevent silent crashes.
- **Bioacoustics Ambient Monitoring:** Directional ceiling mics passively detect acoustic biomarkers (dyspnea, pain) with strict privacy.



Phase 4: Chaos Mitigation

- **Aerospace "Combat Mode":** Triggers under extreme nurse stress to strip 80% of screen data, showing only critical safety flags.
- **Hardware-Agnostic Edge Computing:** Local algorithm initiates "Assume the Worst" protocol if cloud servers drop.





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Thank you

